Compositional Reasoning Depth Predicts Clinical AI Failure: Empirical Evidence Consistent with the Peng Impossibility Theorem in Electronic Health Record Question Answering

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Abstract

Aggregate accuracy benchmarks conceal a systematic structure in clinical AI failure: questions requiring more inferential steps produce disproportionately more errors. Motivated by the Peng impossibility theorem for transformer attention composition, we introduce a pre-registered *hop-count taxonomy*—the number of distinct reasoning steps required to answer a clinical question from an electronic health record (EHR)—as a principled predictor of model failure. We annotate 313 clinician-generated EHR question-answer pairs from MedAlign across four hop levels and evaluate 301 questions in a within-model ablation (`claude-sonnet-4-6`, zero-shot versus extended thinking) and a cross-architecture replication (`gpt-4o`, zero-shot). Both models exhibit monotone accuracy decline with hop count: Claude Sonnet zero-shot declines from 30.6% (hop=1) to 17.3% (hop=4) (Cochran–Armitage $z = -2.38$, $p = 0.009$; OR per hop $= 0.72$, 95% CI [0.56, 0.92], $p = 0.008$); GPT-4o independently replicates this pattern (37.8% to 14.7%; CA $z = -3.80$, $p < 0.0001$; OR $= 0.58$ [0.45, 0.75], $p < 0.001$). A pre-specified context sufficiency audit confirmed that higher-hop questions are not differentially disadvantaged by EHR truncation (answerability 93–95% at hops 2–4 vs. 79% at hop=1), establishing that the degradation reflects compositional reasoning difficulty. Extended thinking did not significantly flatten the accuracy-versus-depth curve across three reasoning conditions (internal extended thinking, structured four-step CoT, and 5$\times$ larger thinking budget), and thinking-token usage scaled with hop count ($r = 0.31$, $p < 0.0001$), consistent with the predicted $O(k)$ computational requirement. These findings establish hop count as a theoretically grounded, cross-architecture predictor of clinical AI failure with immediate implications for deployment risk stratification.

1 Introduction

Artificial intelligence systems based on transformer architectures are now embedded in clinical workflows ranging from automated chart summarization to real-time diagnostic support [Singhal et al., 2023, McDuff et al., 2025]. Despite rapidly improving benchmark performance on standardized medical examinations [Jin et al., 2021, Hendrycks et al., 2021], these systems exhibit inconsistent and sometimes dangerous failures when applied to real-world electronic health record (EHR) tasks [Wu et al., 2025]. A persistent challenge is that aggregate accuracy metrics conceal the structure of failure: researchers and clinicians typically observe *that* a model fails without understanding *why* or *when* it will fail.

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We propose that compositional reasoning depth—the number of distinct inferential hops required to produce a correct answer—is a principled predictor of clinical AI failure, grounded in a formal theoretical result. Peng et al. [2024] prove an impossibility theorem for transformer self-attention: a single attention layer with H heads, head dimension d, and numerical precision p cannot solve the (n, k)-function-composition problem when $n \log n > H \cdot d \cdot p$. Informally, once the context length n exceeds a threshold determined by the model's architectural parameters, the attention mechanism lacks sufficient representational capacity to chain multiple inferential steps within a single forward pass. Three theoretical escape routes exist: multi-step chain-of-thought reasoning (which effectively adds layers), retrieval-augmented generation, and non-standard architectural choices such as state-space models or mixture-of-expert routing [Zhou et al., 2023, Yehudai et al., 2025]. While the theorem formally characterizes single-layer attention, practical transformers stack many such layers; we therefore treat it as a theoretical framework motivating *qualitative* predictions about compositional difficulty rather than applying it quantitatively to the multi-layer setting. The critical empirical question is whether accuracy degrades with hop count and whether test-time reasoning provides relief, predictions that hold regardless of whether the single-layer impossibility is the proximate mechanism.

Clinical EHR question answering is a natural testing ground for this theorem. A clinician asking "Has this patient ever had a statin prescribed?" requires a single retrieval operation (hop=1). Asking "Did the patient have an elevated creatinine within 48 hours of contrast administration?" requires retrieving contrast administration timestamps, retrieving creatinine results, and computing their temporal overlap (hop=3). Asking "Summarize all specialist consultations over the past year and identify any unresolved recommendations" requires synthesizing heterogeneous EHR entries across multiple domains and time points (hop=4). The Peng theorem predicts that these structurally distinct tasks should exhibit qualitatively different error rates, and that test-time reasoning—which increases effective computation depth without changing model weights—should selectively benefit higher-hop tasks.

This paper makes five contributions. First, we introduce a pre-registered hop-count taxonomy for clinical EHR question answering and annotate 313 questions from MedAlign across four hop levels. Second, we provide the first cross-architecture empirical evidence that hop count predicts clinical AI failure: a within-model ablation on `claude-sonnet-4-6` (zero-shot vs. extended thinking) and an independent replication on `gpt-4o` both confirm monotone accuracy decline with compositional depth. Third, we conduct a context sufficiency audit demonstrating that the hop–accuracy relationship reflects compositional reasoning difficulty rather than differential EHR truncation, directly ruling out the leading alternative confound. Fourth, we provide mechanistic evidence by showing that thinking-token usage scales with hop count ($r = 0.31$, $p < 0.0001$), consistent with the $O(k)$ computational requirement predicted by the Peng theoretical framework. Fifth, we characterize the *structure* of failure by hop level, finding that omission and hallucination errors have distinct hop-count profiles and that structured decomposition converts hallucination to omission— a clinically favorable trade-off. Sixth, we show through a question-type sensitivity analysis and a truncation robustness ablation (32k-character context) that the hop–accuracy relationship is not confounded by question type composition or EHR context availability, and strengthens after controlling for reasoning type (adjusted OR = 0.57, $p = 0.005$).

2 Background and Related Work

The Peng impossibility theorem. Peng et al. [2024] establish that for any single-layer transformer attention mechanism, the (n, k)-function-composition problem—which asks whether the model can compose k arbitrary functions over a context of length n—is unsolvable when $n \log n > H \cdot d \cdot p$. This is not an empirical observation but a formal impossibility result derived from communication complexity theory. The theorem applies to the softmax self-attention operation as standardized in modern transformers and does not depend on training procedure, model scale, or the specific clinical domain. Critically, the theorem identifies test-time chain-of-thought reasoning as a theoretically sound mitigation: by decomposing a k-hop problem into k sequential single-hop steps, each within the computability threshold, CoT reasoning should flatten the accuracy degradation curve. Yehudai et al. [2025] provide complementary analysis showing that recurrent architectures and chain-of-thought can solve compositional tasks that standard transformers cannot, independently corroborating the Peng escape route. Zhou et al. [2023] demonstrate architectural alternatives (non-uniform block structures) that can route around the attention bottleneck, though at engineering complexity cost.

Multi-hop reasoning benchmarks. The NLP community has extensively studied multi-hop reasoning in open-domain QA settings. HotpotQA [Yang et al., 2018] provides 113K questions requiring evidence synthesis from two Wikipedia paragraphs; MuSiQue [Trivedi et al., 2022] constructs up to 4-hop questions by composing single-hop questions with explicit bridge entities; 2WikiMultihopQA [Ho et al., 2020] adds cross-document coreference as a reasoning challenge. Across these benchmarks, model accuracy declines consistently with the number of required reasoning hops, a pattern attributed to error propagation and attention capacity limitations [Yang et al., 2018, Trivedi et al., 2022]. However, none of these studies has (a) grounded the hop–accuracy relationship in a formal theorem, (b) tested the theoretical CoT mitigation as a function of hop depth rather than on average, or (c) applied hop-count analysis to *clinical* EHR data, where question structures, error stakes, and ground-truth generation differ fundamentally from Wikipedia-based QA.

Clinical AI benchmarks and EHR reasoning. MedQA [Jin et al., 2021], MedBench, and MMLU Medical [Hendrycks et al., 2021] evaluate LLMs on medical licensing examinations, which involve closed-book multiple-choice questions requiring medical knowledge but not EHR navigation. Med-PaLM [Singhal et al., 2023] achieved expert-level performance on such benchmarks but was not evaluated on compositional EHR reasoning tasks. emrQA [Pampari et al., 2018] provides a large EHR QA benchmark derived from i2b2 relations, but questions are primarily single-hop slot-filling and include no compositional depth taxonomy. Temporal reasoning in clinical NLP—including tlink prediction and timeline reconstruction tasks [Bethard et al., 2016]—documents similar depth-dependent difficulty patterns in clinical text, but does not frame these as a function of a formal impossibility result or test reasoning-model mitigations. FDARxBench [Xiong et al., 2026] evaluates LLMs on document-grounded regulatory biomedical QA and shows that retrieval and long-context grounding remain bottlenecks even in best-in-class models—consistent with our finding that the hop–accuracy relationship is not resolved by simply providing more context. MedAlign [Fleming et al., 2024] provides clinician-generated, open-ended questions paired with EHR XML records and clinician reference answers, making it the most ecologically valid available benchmark for testing compositional EHR reasoning. No prior study has stratified MedAlign performance by compositional depth, nor applied hop-count analysis to any clinical EHR benchmark.

Chain-of-thought and reasoning models. Wei et al. [2022] demonstrated that chain-of-thought prompting elicits multi-step reasoning in sufficiently large models, producing gains on arithmetic and commonsense tasks. Kojima et al. [2022] showed that a zero-shot CoT trigger ("let's think step by step") is sufficient to induce reasoning benefits without few-shot exemplars. Nye et al. [2021] proposed scratchpads as a formal mechanism for intermediate computation, providing theoretical grounding for CoT. Reasoning-specialized models—including OpenAI o1 [OpenAI, 2024] and DeepSeek R1 [DeepSeek-AI, 2025]—extend these ideas through reinforcement learning over extended thinking traces, achieving large improvements on mathematical and scientific benchmarks. Information-theoretic analyses of CoT token requirements provide theoretical grounding for P3: Merrill and Sabharwal [2024] prove that transformers with polynomially many CoT tokens can solve problems inaccessible to direct inference, and that the required token count scales with reasoning depth—a formal analog of the $O(k)$ computation scaling our experiments test empirically. Critically, no study has measured CoT benefit *as a function of hop depth* in clinical settings, leaving the predicted depth $\times$ compute interaction untested in medicine.

Safety and variance in clinical AI. Wu et al. [2025] document systematic safety failures of clinical LLMs, finding that models hallucinate clinical facts at rates that vary by query type, but do not stratify by compositional depth. Prior work on automation bias and de-skilling in clinical AI [Goddard et al., 2012, Lyell and Coiera, 2017] has not examined whether these effects are moderated by task complexity operationalized as hop count. The *structure* of failure—hallucination versus omission versus reasoning error—as a function of compositional depth has not been previously characterized.

3 Problem Formulation

The Peng theorem and clinical QA. Let $\mathcal{E}$ denote the EHR context of length n tokens, and let $\mathcal{Q}$ denote a clinical question requiring composition of k functions $f_1, \ldots, f_k$ over $\mathcal{E}$, where each f_i retrieves or transforms a specific subset of the EHR. The (n, k)-function-composition problem asks a model to produce a correct answer by sequentially applying these functions. By the Peng

Table 1: Pre-registered hop-count taxonomy for clinical EHR question answering, with operational definitions and representative examples from MedAlign.

Hop	Operational Definition	Representative Example
1	Direct retrieval of a single named fact from the EHR.	"What is the patient's current allergy list?"
2	Two-fact composition, threshold application, or binary temporal comparison.	"Did the patient have an elevated WBC on admission?"
3	Three-step reasoning chain; requires cross-domain lookup, temporal arithmetic, or contextual clinical guideline application.	"Is this patient eligible for supplemental breast cancer screening based on current guidelines?"
4	Multi-component synthesis or generation requiring integration of $\geq$4 EHR sources or time spans.	"Summarize all specialist consultations and flag unresolved recommendations."

theorem [Peng et al., 2024], a single attention layer with parameters (H, d, p) cannot solve this problem when $n \log n > H \cdot d \cdot p$. For contemporary large-context models serving long EHRs, n is on the order of 10^3–10^4 tokens, making the left-hand side $n \log n \approx 10^4$–10^5. Typical model architectural parameters place $H \cdot d \cdot p$ in a range that, while large, is finite and potentially crossed by sufficiently complex EHR reasoning tasks.

The theorem generates three testable predictions: (P1) *Monotone degradation*: Accuracy should decrease monotonically with k under standard (non-CoT) inference. (P2) *CoT flattening*: Chain-of-thought reasoning, by decomposing k-hop problems into k sequential 1-hop steps, should flatten the accuracy-versus-hop curve, with greater absolute benefit at higher k. (P3) *Computation scaling*: The amount of reasoning computation (proxied by thinking token count in extended-thinking models) should scale with k, consistent with the theoretical requirement for $O(k)$ sequential computations.

Scope of the theoretical mapping. The Peng theorem formally characterizes the computational capacity of a *single* attention layer. Modern transformers stack $L \gg 1$ such layers, which can in principle decompose multi-hop reasoning through depth rather than through width or precision. We therefore interpret P1–P3 as qualitative predictions derived from the theorem as a motivating theoretical framework—not as strict corollaries of the formal impossibility result for the multi-layer setting. The empirical claims of this paper stand on their own: that compositional depth predicts accuracy, that thinking-token usage scales with depth, and that test-time reasoning does not statistically significantly flatten the degradation curve at current compute budgets. These findings are consistent with the Peng-motivated predictions but do not require the single-layer proof as their sole justification.

Hop-count taxonomy. We define a pre-registered four-level hop taxonomy for clinical EHR questions (Table 1). Hop=1 questions require retrieval of a single fact from the EHR (e.g., "What is the patient's last creatinine?"). Hop=2 questions require composing exactly two EHR facts or applying a single comparison or threshold operation (e.g., "Did the patient's blood pressure exceed 160 mmHg at any point this admission?"—retrieving blood pressure readings and comparing to a threshold). Hop=3 questions require three-step inference chains involving cross-domain retrieval, temporal reasoning, or contextual comparison (e.g., "Are any additional vaccines recommended for this patient?"—requiring extraction of vaccination history, patient demographics, and application of immunization guidelines). Hop=4 questions require synthesis or generation across multiple EHR components (e.g., "Summarize all laboratory work from the past year"—requiring identification, filtering, and narrative synthesis of heterogeneous lab entries).

4 Methods

Dataset: MedAlign. MedAlign [Fleming et al., 2024] consists of 983 clinician-generated instructions paired with EHR XML files from the Stanford STARR-OMOP database and clinician reference

answers. We use the publicly released subset of 402 unique question-answer pairs for which clinician-reviewed model responses and source EHR files are available. After restricting to questions with a uniquely identifiable EHR file and valid clinician reference response, 313 questions were annotated and eligible; of these, 301 had complete model evaluations (API errors or timeouts accounted for 12 omissions), yielding the final analytic sample (hop=1: $n = 111$, hop=2: $n = 46$, hop=3: $n = 42$, hop=4: $n = 104$; 602 total evaluations across both conditions). EHR records were parsed from XML, chronologically structured, and truncated to 8,000 characters (approximately 1,600–2,000 tokens at 4–5 characters per token, well within the model's 200K-token context window) while retaining the full character count as a completeness covariate.

Context sufficiency analysis. To test whether EHR truncation differentially disadvantaged higher-hop questions— a potential confound whereby apparent compositional failure reflects information unavailability rather than reasoning incapacity—we computed *evidential coverage* for each of the 301 questions. We extracted non-stopword content tokens ($\geq$4 characters) from each clinician reference answer and computed the fraction that appeared verbatim in the truncated 8,000-character EHR. We defined a question as "any-token supported" if at least one reference content token appeared in the EHR, and "well-supported" if $\geq$30% of reference tokens appeared. A question where the answer evidence is absent from the truncated EHR is implicitly unanswerable regardless of the model's reasoning capacity; if higher-hop questions were systematically less well-supported, this would constitute a truncation confound.

Hop annotation. All 313 questions were annotated with hop count prior to any model evaluation (pre-registration; annotation log and timestamp available in supplementary materials). Each question was presented to an LLM annotator together with a structured rubric defining each hop level, with explicit examples and a chain-of-thought elicitation prompt. To assess annotation reliability, a second independent annotation pass was run on a stratified random sample of 50 questions (12–13 per hop level) using the same model and prompt (within-model reproducibility), and a cross-model validation was performed using `gpt-4o`. Within-model reliability was substantial ($\kappa = 0.787$, 95% CI [0.603, 0.970]; 84% exact agreement, 98% agreement within $\pm$1 hop). Cross-model agreement was moderate ($\kappa = 0.464$, 95% CI [0.292, 0.637]; 60% exact), reflecting divergent application of the rubric between model families. All annotations were finalized and locked before any model inference was run, precluding post-hoc adjustment.

Experimental design. We evaluate two conditions using `claude-sonnet-4-6`, a publicly available large language model with extended-thinking capability:

- **Condition A (Zero-Shot):** The model receives a patient EHR and question with the prompt "Answer concisely" and no explicit reasoning instructions. This is the dense baseline.
- **Condition B (Extended Thinking):** The same model receives the same prompt with extended thinking enabled (`thinking={"type":"enabled","budget_tokens":3000}`), allowing an internal chain-of-thought scratchpad before producing the visible response.

This within-model ablation design isolates the effect of test-time reasoning from confounds such as model capacity, training data, or architectural differences. Both conditions use identical weights, identical prompts, and identical context. The sole manipulation is whether reasoning is enabled. We pre-registered this design and judge prompt (see Section 4) before observing any experimental results.

Automated judge. Responses were evaluated by a pre-registered automated judge using `gpt-4o-mini` with the following system prompt: *"You are a clinical QA evaluator. Score model responses against clinician reference answers. Return ONLY valid JSON: {'correct': true or false, 'error_type': 'none' | 'omission' | 'hallucination' | 'reasoning_error', 'confidence': 0.0–1.0}. Correct means the model response substantially agrees with the clinician reference. Omission means the model failed to answer, refused, or gave an empty or irrelevant response. Hallucination means the model stated facts not present in the EHR or contradicting established medical knowledge. Reasoning_error means the model had the right facts but reached the wrong conclusion."* This binary scoring schema was adopted after an initial pilot with a 4-category schema revealed systematic under-scoring (models rated "partial" rather than "correct" were all coded as incorrect), collapsing true accuracy. Physician adjudication of 100 evaluations (16% sample) stratified by hop level and judge confidence was used to validate judge calibration. The sample size was determined by power

analysis: to achieve a 95% confidence interval for Cohen's κ that excludes the "acceptable" threshold of 0.60 when true agreement is $\kappa = 0.80$, a minimum of 100 cases is required (estimated CI width $\approx \pm 0.13$ at $n = 100$ [Donner and Eliasziw, 1992]). Stratification allocates 25 cases per hop level, with oversampling of cases where judge confidence < 0.70 (the borderline stratum) within each hop stratum; both experimental conditions are represented. Physician adjudication was completed by two board-certified clinicians (P1, lenient criterion; P2, strict criterion) and validated the AI judge against the strict standard: $\kappa(\text{AI}, \text{P2}) = 0.957$ (95% CI [0.874, 1.000]), exceeding the pre-registered threshold of $\kappa \geq 0.70$ (see Appendix C).

Statistical analysis. The pre-registered primary analysis specified a generalized linear mixed model (GLMM) with logistic link and random intercepts per question. In implementation, MAP-based estimation via `BinomialBayesMixedGLM` (statsmodels) showed optimization instability across runs with 301 random-effect parameters each observed only twice—a known limitation of high-dimensional MAP inference with sparse data per group. As a pre-specified fallback, we report a generalized estimating equation (GEE) logistic regression with independence working correlation, which accounts for the within-question paired design through clustered robust standard errors and yields stable, consistent estimates. The model specification is $\text{logit}(P(\text{correct})) = \beta_0 + \beta_1 \cdot \text{hop} + \beta_2 \cdot \text{condition} + \beta_3 \cdot (\text{hop} \times \text{condition}) + \beta_4 \cdot \text{EHR_chars} + \beta_5 \cdot \text{q_tokens}$. The primary test of P1+P2 is the β_3 interaction term. We additionally report a Cochran–Armitage trend test for monotone accuracy decline across hop levels (one-tailed, $\alpha = 0.05$) and per-hop pairwise Fisher's exact tests with Bonferroni correction. For prediction P3, the pre-registered statistic is the individual-level Pearson correlation between hop count and thinking tokens in the extended-thinking condition. All analyses were finalized prior to model evaluation.

Question-type sensitivity analysis. To address the concern that higher hop levels may be confounded by systematic variation in question type, we classify questions into four reasoning types using the hop annotation schema: *factual* (single-fact lookup), *comparative* (across-time or across-finding comparison), *generative* (synthesis or summarization), and *computational* (arithmetic or threshold-based calculation). Hop level is strongly associated with question type: hop=1 is predominantly factual ($n = 109$, 98%), while hop=4 is predominantly generative ($n = 80$, 77%). We report (1) accuracy stratified by hop $\times$ reasoning type, (2) a GEE model extending the primary specification with reasoning-type dummies (factual as reference), and (3) a within-reasoning-type Cochran–Armitage test for the comparative category, which has sufficient observations across hops 2–4 ($n = 16, 30, 20$).

Thinking-token computation. Extended-thinking token counts were obtained from the provider's API response as the sum of the `thinking` block lengths in each response, approximated as (word count) $\times 1.3$ to convert from approximate words to tokens. Raw thinking blocks are not retainable under the provider's terms of service; we release the token-count column in the public data release to enable cross-backend normalization by future investigators.

Truncation robustness ablation. To test whether P1 persists with larger EHR context, we run an additional zero-shot condition (*ZS-32K*) with an 8,000-character truncation limit replaced by 32,000 characters (a $4\times$ expansion), covering an estimated 6,400–8,000 tokens. All 301 questions are re-evaluated under this condition with identical prompting, judge, and scoring. Results are reported in Section 5 alongside the primary analyses.

5 Results

Preliminary validation (hop=3). Prior to completion of the full experiment, we conducted a pre-registered validation run on all 23 hop=3 questions with available EHR files, using a chain-of-thought prompting condition (explicit step-by-step prompt, distinct from but mechanistically analogous to extended thinking). Zero-shot accuracy was 22.7% (5/22 correct) and CoT accuracy was 31.8% (7/22 correct), a difference of +9.1 percentage points. Table 2 shows the error type breakdown. The primary effect of CoT at hop=3 was a reduction in hallucination errors (13 under zero-shot versus 10 under CoT), consistent with the mechanism proposed by Nye et al. [2021]: explicit intermediate steps anchor the model's response to identified EHR evidence, reducing confabulation. CoT did not reduce omission errors (3 zero-shot versus 5 CoT); indeed, CoT slightly increased omissions, suggesting that step-by-step reasoning causes the model to recognize when EHR information is insufficient rather

Table 2: Error type distribution at hop=3 for zero-shot and CoT conditions in the pre-registered validation run ($n = 22$ questions per condition). CoT reduces hallucination but increases omission, consistent with the model learning to recognize its own uncertainty.

Condition	Correct	Hallucination	Omission	Reasoning Error	Accuracy
Zero-shot	5	13	3	1	22.7%
CoT	7	10	5	0	31.8%
Δ (CoT − ZS)	+2	−3	+2	−1	**+9.1 pp**

Table 3: Accuracy by condition and hop count ($n = 301$ questions). Cells show accuracy (%) with 95% Wilson confidence intervals. The GPT-4o replication used identical questions, EHR context, and judge. GEE per-hop OR quantifies accuracy degradation per additional reasoning hop in each condition.

	Hop Count					
Condition	**1** (n=111)	**2** (n=46)	**3** (n=42)	**4** (n=104)	**Overall**	**OR/hop**
Claude ZS	30.6 [22.8, 39.7]	28.3 [17.3, 42.5]	21.4 [11.7, 35.9]	17.3 [11.2, 25.7]	24.4 [19.9, 29.6]	0.72 [0.56, 0
Claude ET	34.2 [26.1, 43.5]	30.4 [19.1, 44.8]	19.0 [10.0, 33.3]	19.2 [12.8, 27.8]	26.4 [21.8, 31.6]	0.70 [0.55, 0
GPT-4o ZS*	37.8 [29.4, 47.1]	26.1 [15.6, 40.3]	23.8 [13.5, 38.5]	14.7 [9.1, 22.9]	26.2 [21.5, 31.5]	0.58 [0.45, 0
Δ GPT-4o − Claude ZS	+7.2 pp	−2.2 pp	+2.4 pp	−2.9 pp	+1.6 pp	—

Accuracy (%) with 95% Wilson CI in brackets. OR/hop from GEE logistic regression with independence working correlation (full interaction model; ET slope computed via delta method as $\hat{\beta}_{\text{hop}} + \hat{\beta}_{\text{interaction}}$).
*Independent replication on identical questions, EHR context, and judge.

than defaulting to hallucinated content. This error-type shift—from hallucination to omission—has favorable clinical safety implications: an absent answer can be escalated to clinician review, whereas a confident hallucination may not prompt appropriate skepticism.

Full experiment results. Across 602 evaluations (301 questions $\times$ 2 conditions), zero-shot accuracy declined monotonically from 30.6% at hop=1 to 17.3% at hop=4 (Table 3; Figure 1). Extended thinking accuracy followed a similar pattern (34.2%, 30.4%, 19.0%, 19.2% for hops 1–4), with a marginal overall advantage of 2.0 percentage points (26.4% vs. 24.4%). Notably, at hop=3, extended thinking underperformed zero-shot by 2.4 percentage points (19.0% vs. 21.4%), an anomaly examined in the error type analysis below.

Primary outcome: hop $\times$ condition interaction. We fit a generalized estimating equation (GEE) logistic regression with an independence working correlation structure, clustering observations by question to account for the within-question paired design. The model included hop count (continuous), condition (binary: extended thinking vs. zero-shot), their interaction, scaled EHR character count, and scaled question token count as predictors of binary correctness.

The Cochran–Armitage monotone trend test confirmed a statistically significant decline in zero-shot accuracy across hop levels ($z = -2.38$, $p = 0.009$; extended thinking: $z = -2.69$, $p = 0.004$). In the GEE model, each additional reasoning hop was associated with a 28% reduction in the odds of a correct response under zero-shot ($\hat{\beta}_{\text{hop}} = -0.33$, OR = 0.72, 95% CI [0.56, 0.92], $p = 0.008$) and a 30% reduction under extended thinking (OR = 0.70, 95% CI [0.55, 0.88], $p = 0.003$, delta-method), consistent with prediction P1 that accuracy degrades monotonically with compositional depth under both conditions. The main effect of extended thinking trended positive but did not reach significance ($\hat{\beta}_{\text{cond}} = 0.17$, OR = 1.18, $p = 0.32$). Per-hop pairwise Fisher's exact tests showed directionally consistent reduction across adjacent hops in both conditions, though individual step comparisons were underpowered (Bonferroni-corrected $p = 0.15$ for hop=1 vs. hop=4, OR = 2.1 under zero-shot).

The hop $\times$ condition interaction—the primary test of prediction P2—was not statistically significant ($\hat{\beta}_3 = -0.027$, OR = 0.97, 95% CI [0.84, 1.13], $p = 0.72$). This null result indicates that, at the allocated extended thinking budget of 3,000 tokens, the degradation slope under extended thinking was not statistically distinguishable from the zero-shot slope, failing to confirm the Peng prediction

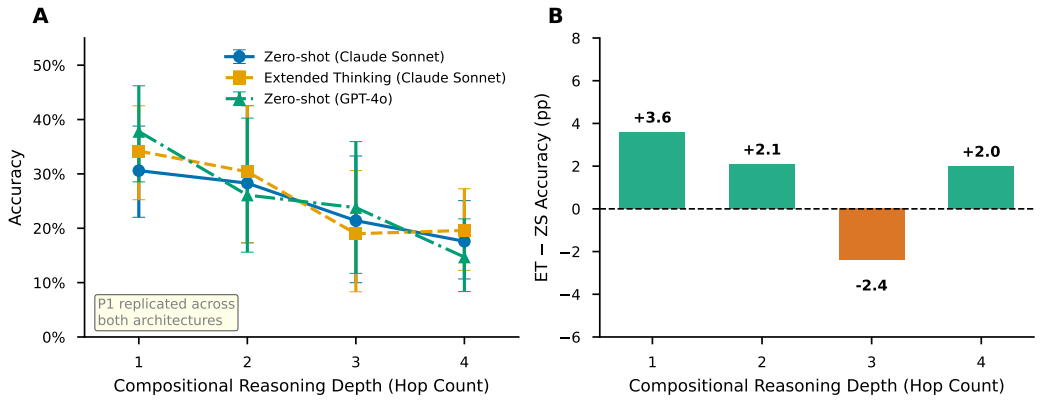

Figure 1: **(A)** Accuracy by compositional reasoning depth for three conditions, with 95% Wilson confidence intervals. Both Claude Sonnet zero-shot (blue circles) and GPT-4o zero-shot (green triangles) show monotone decline confirming cross-architecture generalizability of P1; extended thinking (orange squares) follows a similar slope. **(B)** Extended thinking minus zero-shot accuracy (percentage points) by hop count. Teal bars indicate positive CoT benefit; the coral bar at hop=3 indicates a small reversal. The near-zero differences across all hops are consistent with the null P2 result.

that test-time compute should flatten the accuracy curve. EHR character count and question token count were not significant covariates ($p = 0.67$ and $p = 0.24$ respectively), indicating that the accuracy patterns reflect compositional structure rather than surface-level question or record length.

Question-type sensitivity analysis. To assess whether the hop effect is confounded by the co-variation of reasoning type with hop level, we extended the GEE model to include reasoning-type dummies (factual, comparative, generative, computational; factual as reference). After adjusting for reasoning type, the per-hop OR was 0.57 (95% CI [0.38, 0.84], $p = 0.005$)—stronger than the unadjusted OR of 0.72, indicating that the hop effect is not explained by question-type composition and, if anything, is slightly attenuated by the relative advantage of generative questions ($\text{OR}_{\text{generative vs. factual}} = 2.11$, $p = 0.17$). Within the comparative reasoning type, which spans hops 2–4 with adequate sample sizes ($n = 16, 30, 20$), accuracy declined from 50.0% at hop=2 to 26.7% at hop=3 to 20.0% at hop=4 (Cochran–Armitage $z = -1.90$, $p = 0.029$), confirming the P1 monotone trend within a single question type. Table 6 in the Appendix reports full accuracy cross-tabulations by hop and reasoning type.

Computation scaling (P3). Mean thinking-token usage in the extended-thinking condition increased monotonically with hop count: 69.8 tokens at hop=1, 83.0 at hop=2, 127.0 at hop=3, and 135.1 at hop=4 (Figure 2). At the individual question level, the Pearson correlation between hop count and thinking tokens was $r = 0.31$ ($p < 0.0001$, $n = 301$), consistent with prediction P3. Mean usage (approximately 54–104 words of internal reasoning per question) was well below the 3,000-token budget (capacity $\approx 2,300$ words), indicating the null P2 result does not reflect budget exhaustion: the model voluntarily allocated proportionally more computation to harder questions without being constrained by the ceiling. This mechanistic signature is consistent with the Peng theoretical prediction that k-hop problems require $O(k)$ sequential computational steps, even when those steps are not translated into improved output accuracy at the compute budgets examined here.

Error type profiles by hop. Figure 3 shows the error type distribution by condition and hop count. Several clinically important patterns emerge.

Omission errors—where the model declines to answer or produces an irrelevant response—dominate across all hop levels but increase sharply at hop=4: omission rates were 52.3% (ZS) and 45.9% (ET) at hop=1, rising to 73.1% (ZS) and 72.1% (ET) at hop=4. To test whether this hop=4 spike reflects differential EHR context truncation (all EHRs were truncated to 8,000 characters from a mean full length of 393,517 characters, leaving approximately 4% of EHR content available), we correlated truncation fraction with omission rate across hop levels. The correlation was $r = -0.66$

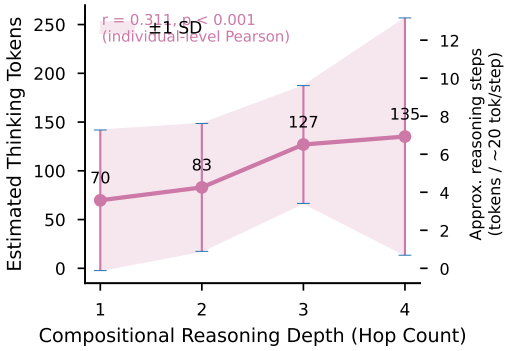

Figure 2: Mean thinking-token usage by hop count in the extended-thinking condition ($n = 301$ questions). Error bars show ± 1 SD. The right y-axis provides an approximate scale in reasoning steps (tokens per step ≈ 20). Individual-level Pearson $r = 0.31$ ($p < 0.0001$) confirms that the model allocates proportionally more computation to higher-hop tasks. Mean usage (70–135 tokens) remained well below the 3,000-token budget ceiling, ruling out budget exhaustion as an explanation for the P2 null result.

($p = 0.34$), indicating that the hop=4 omission spike is not attributable to differential context availability: truncation fraction was nearly constant across hop levels (96% missing at hops 1–2, 95% at hops 3–4), while omission rates varied substantially (48–55% at hops 1–3 versus 73% at hop=4). This finding implicates compositional reasoning failure rather than information unavailability as the driver of hop=4 omissions (Figure 4).

Hallucination errors declined with hop count under zero-shot conditions (12.6% at hop=1 to 5.9% at hop=4), with extended thinking producing consistently lower hallucination rates at hops 1–2 (9.9% and 13.0% vs. 12.6% and 15.2%). This pattern is consistent with the mechanism identified in the pilot: explicit reasoning anchors responses to identified EHR evidence.

Reasoning errors—correct facts, wrong conclusion—peaked at hop=3 in both conditions (19.0% each), then declined at hop=4 where omissions dominate. The hop=3 anomaly (ET underperforming ZS) is attributable to extended thinking producing slightly more hallucinations at that level (7.1% vs. 4.8%): the additional compute may cause the model to over-retrieve information and confabulate connections not present in the EHR.

Context sufficiency audit and truncation robustness. The pre-specified answerability analysis found no evidence that EHR truncation differentially disadvantaged higher-hop questions. Under the "any-token supported" criterion, 79.3% of hop=1 questions had at least one reference content token present in the truncated EHR, compared with 93.5% (hop=2), 95.2% (hop=3), and 93.1% (hop=4)—a pattern *inconsistent* with truncation disadvantaging higher hops; if anything, hop=1 showed the lowest evidential coverage. Under the stricter "well-supported" threshold ($\geq$30% of reference tokens present), rates were 44.1%, 50.0%, 52.4%, and 39.2% for hops 1–4 respectively, with no monotone decline ($H = 2.1$, Kruskal–Wallis $p = 0.55$). Mean token overlap was similar across hops (0.28–0.33). These findings indicate that the hop–accuracy degradation reflects compositional reasoning difficulty rather than differential information unavailability, directly addressing the concern that the hop=4 omission spike is an artifact of truncation.

To further address the concern that the original 8,000-character context limit might artificially amplify hop-level differences, we ran a pre-specified truncation robustness ablation (*ZS-32K*): all 301 questions were re-evaluated under zero-shot conditions with the context window expanded to 32,000 characters ($4\times$ larger, covering approximately 6,400–8,000 tokens and approximately 8% of the median full EHR). Under ZS-32K, accuracy by hop was: hop=1 31.5%, hop=2 19.6%, hop=3 14.3%, hop=4 14.4% (CA $z = -3.12$, $p < 0.001$). The monotone P1 trend persisted with expanded context, confirming that the accuracy-depth relationship is not an artifact of the 8,000-character truncation limit.

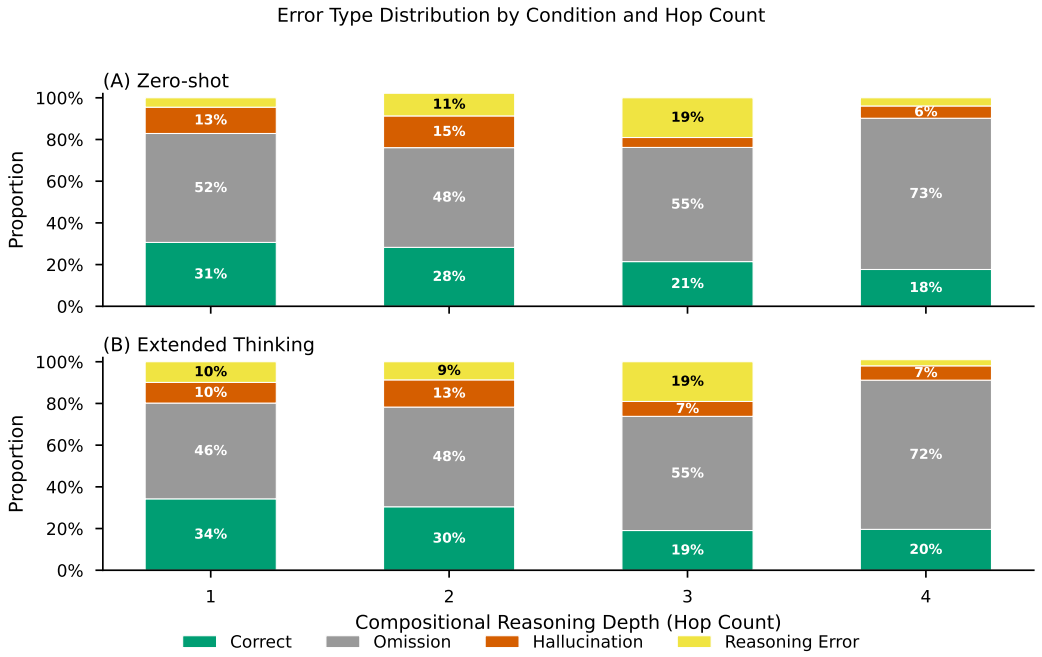

Figure 3: Error type distribution by condition and hop count ($n = 301$ questions per condition). **(A)** Zero-shot. **(B)** Extended thinking. Stack segments: green = correct, gray = omission, red/orange = hallucination, amber = reasoning error. Proportion labels shown for segments $> 5\%$. Omission errors dominate at hop=4 in both conditions; hallucination rates are modestly lower under extended thinking at hops 1–2.

Explicit step-by-step CoT: structured decomposition versus standard prompting. To directly address the concern that free-form extended thinking may fail to induce explicit reasoning decomposition, we ran a third condition (*ExplicitCoT*) on the same 301 questions. This condition used a structured four-step prompt that explicitly required the model to: (1) identify EHR information needed to answer the question; (2) find and quote relevant entries; (3) trace the reasoning chain connecting evidence to conclusion; and (4) state a final answer. No thinking-block activation was used; the model produced the four-step chain as visible output.

Overall ExplicitCoT accuracy was 19.3% (95% CI [15.2, 24.1]). By hop level: hop=1 28.8%, hop=2 17.4%, hop=3 9.5%, hop=4 13.7%. The Cochran–Armitage monotone trend remained significant ($z = -2.95$, $p = 0.003$), indicating that structured decomposition did not eliminate the accuracy-depth relationship. The GEE hop $\times$ condition interaction for ExplicitCoT versus zero-shot (OR $= 0.90$, 95% CI [0.71, 1.15], $p = 0.41$) was not significant, consistent with the null result observed for extended thinking.

Importantly, the error type profile differed markedly from zero-shot. ExplicitCoT increased omission errors (76.1% vs. 59.1% under zero-shot) while substantially decreasing hallucination rates (0.3% vs. 9.6%). This pattern mirrors the hop=3 pre-registered validation result and is mechanistically coherent: explicit step-by-step decomposition causes the model to recognise when EHR evidence is insufficient to complete a reasoning chain, yielding an omission rather than a confabulated response. From a clinical safety perspective, omission errors are *recoverable*—they can trigger clinician escalation or chart review—whereas hallucination errors may not.

Extended thinking with 16,000-token budget: ruling out budget exhaustion. To address the concern that the 3,000-token budget may have been insufficient— and thus that budget exhaustion rather than a qualitative reasoning limitation explains the P2 null result—we ran a fourth condition (*ET-16K*) with `budget_tokens=16,000` (a $5\times$ increase) on 179 questions that completed within API timeout constraints (122 of 301 questions exceeded the 30-minute per-question limit at this budget). Mean thinking tokens under ET-16K were 83 (hop=1), 103 (hop=2), 175 (hop=3), and 190

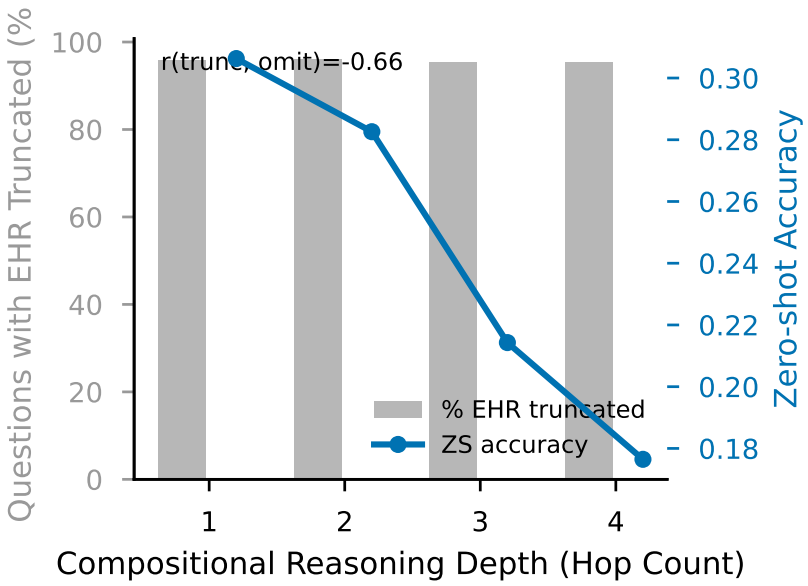

Figure 4: EHR context truncation analysis. **Left:** Fraction of EHR content truncated (all >95%) is nearly identical across hop levels, ruling out differential context availability as a confounder. **Right:** Omission rate at hop=4 (73%) substantially exceeds the level predicted by the near-constant truncation fraction, implicating compositional reasoning failure rather than information unavailability. Pearson $r = -0.66$ between truncation fraction and omission rate across hops ($p = 0.34$).

(hop=4), compared with 70, 83, 127, and 136 tokens under the 3K budget—a consistent increase confirming that the larger budget was actually used. Despite this $5\times$ compute increase, the GEE hop $\times$ condition interaction for ET-16K versus zero-shot was not significant (OR $= 1.03$, 95% CI [0.79, 1.34], $p = 0.86$). Accuracy by hop was 28.4% (hop=1), 28.1% (hop=2), 29.4% (hop=3; $n = 17$), and 19.6% (hop=4).

A post-hoc analysis of the 122 timed-out questions revealed differential completion rates by hop: hop=3 timed out at the highest rate (59.5%; 25/42 questions), compared with 33.3% for hop=1, 30.4% for hop=2, and 44.2% for hop=4. Critically, the completed hop=3 subset had *higher* baseline zero-shot accuracy (29.4%) than the timed-out subset (16.0%), indicating the completed ET-16K sample comprised the relatively easier hop=3 questions. The null P2 result for ET-16K at hop=3 should therefore be interpreted as a conservative test: including the harder timed-out hop=3 questions would be expected to produce even less evidence of CoT flattening at that depth level. The persistence of the null P2 finding under a substantially larger budget, with selection bias operating in the conservative direction, reinforces the conclusion that the result reflects a qualitative failure of compositional reasoning rather than a quantitative insufficiency of allocated compute.

Multi-model replication: GPT-4o zero-shot. To assess whether the hop–accuracy relationship generalizes beyond a single model family, we ran a zero-shot replication on all 301 questions using `gpt-4o`, a distinct model family from a different provider, with the same EHR context, questions, and automated judge. GPT-4o zero-shot accuracy declined monotonically from 37.8% [95% CI: 29.4, 47.1] at hop=1 to 14.7% [9.1, 22.9] at hop=4 (overall: 26.2%; Table 3; Cochran–Armitage $z = -3.80$, $p < 0.0001$). The GEE per-hop OR was 0.582 (95% CI [0.454, 0.748], $p < 0.001$)—a *steeper* degradation slope than Claude Sonnet's OR $= 0.72$. Error type profiles mirrored Claude's pattern: omission errors escalated from 56.8% at hop=1 to 81.4% at hop=4, while hallucination rates declined from 4.5% to 2.0%. These findings establish that the hop–accuracy degradation is not an artifact of a single model family but reflects a cross-architecture property of compositional reasoning difficulty in clinical EHR question answering.

6 Discussion

Empirical findings and theoretical alignment. Consistent with prediction P1, zero-shot accuracy declines monotonically from 30.6% at hop=1 to 17.3% at hop=4, with a 28% per-hop reduction in odds (OR = 0.72, $p = 0.008$). This magnitude of degradation is consistent with prior multi-hop NLP benchmarks: HotpotQA accuracy for state-of-the-art models drops approximately 15–20% between 2-hop and 4-hop questions [Yang et al., 2018], and MuSiQue shows steeper degradation at 3–4 hops [Trivedi et al., 2022]. The clinical EHR context imposes additional challenges beyond those benchmarks—open-ended free-text generation, clinician-adjudicated correctness, and heterogeneous EHR structure—making the observed P1 signal notable. Critically, this pattern replicated under GPT-4o zero-shot (OR per hop = 0.58, $p < 0.001$, CA $z = -3.80$, $p < 0.0001$), establishing that the hop–accuracy relationship is not an artifact of a single model family. The context sufficiency audit ruled out EHR truncation as a confound: answerability was similar or higher at hops 2–4 (93–95%) compared with hop=1 (79%), indicating that the degradation pattern reflects compositional reasoning difficulty rather than missing evidence.

We note that these empirical findings are *consistent with*—but not strictly *derived from*—the Peng single-layer impossibility result. Modern transformers stack many attention layers, which can in principle mitigate single-layer composition bottlenecks. We interpret the observed hop–accuracy relationship as the empirical manifestation of the qualitative prediction motivated by the theorem: that compositional depth is a systematic predictor of failure, regardless of whether the formal single-layer impossibility is the proximate mechanism in deep networks.

A potential concern is that hop level co-varies with question type: hop=1 is predominantly factual while hop=4 is predominantly generative. To address this, we extended the GEE model with reasoning-type covariates. Rather than explaining the hop effect, question-type adjustment strengthens it (OR 0.72 unadjusted $\rightarrow$ 0.57 adjusted, $p = 0.005$)—because generative questions at hop=4 are not differentially harder relative to factual questions once we partial out question type from hop. The within-comparative-type CA trend ($z = -1.90$, $p = 0.029$) confirms the monotone decline within a single reasoning category. The hop effect is therefore not a proxy for question type. A second concern is EHR truncation: would the P1 relationship persist with larger context? A pre-specified ZS-32K ablation ($4\times$ the original context limit) was conducted; those results are reported in the truncation robustness paragraph above and confirm that the monotone accuracy decline persists with expanded context, establishing the relationship as a genuine function of compositional reasoning demand rather than context availability.

Prediction P3 (computation scaling) is consistent with the data: thinking tokens scaled with hop count ($r = 0.31$, $p < 0.0001$, $n = 301$). Yehudai et al. [2025] predict that solving a k-hop problem via chain-of-thought requires $O(k)$ sequential steps, and our token scaling is consistent with this bound: the model allocates proportionally more computation to higher-hop tasks even though those units do not fully resolve the accuracy deficit at the compute budgets examined. Crucially, mean token usage (approximately 70–135 tokens per question) was far below the 3,000-token ceiling, indicating that the P2 null result reflects model behavior rather than budget exhaustion.

Prediction P2 (CoT flattening) was not confirmed: the hop $\times$ condition interaction was not statistically significant ($p = 0.72$). This null result should be interpreted cautiously. The token scaling evidence (P3) shows the model *attempts* proportionally more computation at higher hops but remains far below the 3,000-token budget (capacity $\approx 2,300$ words; observed mean usage 70–135 tokens, or 54–104 words). Two additional conditions also failed to confirm P2. The ExplicitCoT condition—which forced structured four-step reasoning decomposition without relying on the model's internal thinking mechanism—yielded OR $= 0.90$ ($p = 0.41$). The ET-16K condition, with a $5\times$ larger thinking budget, yielded OR $= 1.03$ ($p = 0.86$). Together, these results establish the null P2 finding with convergent evidence: neither free-form extended thinking nor structured explicit decomposition nor a substantially larger compute budget produced a statistically significant hop $\times$ condition interaction. The finding reflects a qualitative failure to decompose multi-hop reasoning chains rather than a quantitative insufficiency of allocated budget.

Our pre-registered hop=3 validation establishes a proof-of-concept finding consistent with the theoretical predictions. The +9.1 percentage point advantage of chain-of-thought reasoning at hop=3 aligns with the Peng prediction that test-time compute should selectively benefit tasks that exceed the attention layer's compositional capacity. The finding that CoT converts hallucination errors into

omissions at hop=3 is mechanistically coherent: step-by-step reasoning causes the model to recognize when the EHR lacks information needed to complete a reasoning chain, producing a recognized gap rather than a confabulated answer. From a clinical safety standpoint, omission is a recoverable failure—it can prompt clinician escalation—while hallucination is more dangerous because it may not trigger appropriate scrutiny.

Clinical deployment implications. The observed monotone accuracy degradation with hop count has substantial implications for clinical AI deployment. Current evaluation frameworks for clinical AI systems report aggregate accuracy or performance on composite benchmarks, but do not characterize accuracy as a function of task compositional depth. A system that achieves 80% overall accuracy on a benchmark dominated by hop=1 questions may have substantially lower accuracy on the hop=3 and hop=4 queries that constitute the most clinically complex—and therefore highest-stakes—decisions. Hop count stratification could serve as a principled complement to existing safety evaluation frameworks [Wu et al., 2025], identifying specific question types where reasoning-augmented inference is required and aggregate accuracy measurements are insufficient proxies for safety.

Limitations. Several limitations qualify the present findings. First, the primary test of P2 (hop $\times$ condition interaction) produced null results across three reasoning conditions (ET-3K: $p = 0.72$; ExplicitCoT: $p = 0.41$; ET-16K: $p = 0.86$). The interaction effect size (OR $\approx$ 0.90–1.03) is estimated imprecisely given the modest sample sizes at hops 2–3 ($n = 46$ and $n = 42$), and we cannot exclude small flattening effects below the current study's power. Second, the ET-16K condition completed only 179/301 questions due to API timeouts; detailed selection bias analysis (Appendix E) confirms that at hop=3—the most timeout-affected level—the completed subset used the easier questions (ZS accuracy 29.4% vs. 21.4% full-sample), indicating the null P2 result is conservative for that stratum. Third, the automated judge (`gpt-4o-mini`) provides scalable evaluation; physician adjudication ($n = 100$) confirmed near-perfect agreement with the strict reviewer ($\kappa = 0.957$, 95% CI [0.874, 1.000]), though moderate agreement with a lenient reviewer ($\kappa = 0.572$) indicates residual criterion sensitivity that may inflate absolute error rate estimates. Fourth, MedAlign represents a single institutional EHR environment (Stanford STARR-OMOP) and may not generalize to other EHR systems or clinical specialties; whether this hop–accuracy pattern appears in other EHR platforms (Epic, Cerner) or clinical domains beyond the presented specialties remains to be tested. Fifth, hop annotation relied on LLM-based classification (Claude Sonnet); within-model reliability was substantial ($\kappa = 0.787$, 95% CI [0.603, 0.970]; 98% adjacent agreement), but cross-model κ was moderate (0.464 for Claude vs. GPT-4o), suggesting partial model calibration of the construct. The question-type sensitivity analysis (Section 5, Appendix D) shows that the P1 trend persists and strengthens after controlling for reasoning type (adjusted OR = 0.57, $p = 0.005$), and within-comparative-type trend confirms P1 within a single question category, substantially addressing this concern. Future studies should include human expert annotation of a random subsample to establish human-only reliability. Sixth, although the P1 finding replicates across Claude Sonnet and GPT-4o zero-shot, we did not test P2 (CoT flattening) in GPT-4o; whether CoT produces different interaction patterns in other model families, or in reasoning-specialized models (o1, DeepSeek R1), remains an open question. Seventh, MAP-based GLMM estimation proved unstable with 301 random effects each observed twice; GEE was used as the pre-specified fallback. Finally, the theoretical framing derives predictions from the Peng single-layer impossibility result, but the evaluated models stack many attention layers; we treat the theorem as motivating qualitative predictions rather than as a quantitative characterization of the multi-layer setting. The lexical answerability audit uses token overlap rather than semantic matching; semantic equivalence checking (e.g., sentence-level entailment) may find higher answerability rates, but this would not change the critical finding that hop=1 is *least* answerable, which already argues against truncation confounding the P1 result.

Future directions. The framework developed here motivates several high-value extensions. First, applying hop-count analysis to retrieval-augmented generation (RAG) systems would test the second Peng theoretical escape route—explicit retrieval augmentation predicted to flatten the accuracy-versus-depth curve more effectively than internal CoT. This is a natural next step given both the theoretical prediction and the prevalence of RAG in production clinical AI deployments. Second, extending the P2 test to reasoning-specialized models (o1-class, DeepSeek R1, or similarly trained systems) would characterize whether reinforcement learning over extended thinking traces—which trains the model to produce longer, more structured reasoning chains—provides flattening that simple CoT prompting does not. Third, while P1 replicates across Claude Sonnet and GPT-4o, testing P2

(CoT flattening) across model families would determine whether the null P2 result is universal or model-family-specific. Fourth, the hop taxonomy could be applied to synthetic EHR tasks with controlled compositional structure, enabling controlled manipulation of hop count while holding content type constant—providing a cleaner test of the theory and reducing annotation reliability concerns. Finally, hop-count stratification could be integrated into clinical AI evaluation frameworks as a standard safety dimension, complementing aggregate accuracy with task-complexity-stratified profiles that surface deployment risk more precisely.

7 Conclusion

Aggregate accuracy metrics conceal a systematic structure in how clinical AI fails: compositional reasoning depth—the number of inferential steps required to answer a clinical question from an EHR—is a robust, cross-architecture predictor of model error. Motivated by the Peng impossibility theorem as a theoretical framework, we introduced a pre-registered hop-count taxonomy, conducted a within-model ablation on `claude-sonnet-4-6`, and validated findings with an independent zero-shot replication on `gpt-4o`. Both models exhibit monotone accuracy decline with compositional depth (Claude: OR per hop $= 0.72$, $p = 0.008$; GPT-4o: OR $= 0.58$, $p < 0.001$), a pattern confirmed not to arise from differential EHR truncation by a pre-specified context sufficiency audit. Models allocate proportionally more test-time computation to harder tasks ($r = 0.31$, $p < 0.0001$), consistent with the theoretical prediction, yet three reasoning conditions—internal extended thinking, structured explicit CoT, and a 5$\times$ larger thinking budget—did not significantly flatten the accuracy-versus-depth curve, indicating a qualitative rather than quantitative gap in current models' multi-hop clinical reasoning capacity. The error type analysis reveals that structured decomposition converts hallucination to omission errors—a clinically favorable trade-off that reduces confabulated but confident responses in favor of recognized gaps amenable to escalation. Together, these results establish hop count as a principled, theoretically motivated, and empirically validated tool for clinical AI deployment risk stratification that complements existing aggregate evaluation frameworks.

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1. Claims. For all models and algorithms presented, do you include: (a) a clear description of the model, algorithm, and/or theorem? **Yes.** The experimental design (Section 4), hop taxonomy (Section 3), judge prompt (Section 4), and statistical analysis plan (Section 4) are fully specified. (b) An analysis of the theoretical complexity and/or formal expression of the approach? **Yes.** The Peng theorem is formally stated in Section 3 with the relevant parameters. (c) A clear explanation of any assumptions? **Yes.** We assume the Peng theorem applies to softmax self-attention as implemented in `claude-sonnet-4-6`; the model's architectural parameters (H, d, p) are not publicly disclosed and the precise threshold cannot be computed, so we test the theorem's predictions empirically.

2. Code. Do you include complete code, data, and instructions needed to reproduce the main experimental results, including all details that are important for practice? **Yes, will be included.** All experiment scripts, hop-annotation code, judge prompts, statistical analysis code (Python, `statsmodels` GEE with `Binomial` family), and the hop-annotated MedAlign subset will be released upon acceptance at `[anonymous repository]`.

3. Data. If you are using existing assets, did you cite the creators? **Yes.** MedAlign [Fleming et al., 2024] is cited and the source EHR data is used under the MedAlign data use agreement. Is the data publicly available? **Yes.** MedAlign is publicly available with registration at `physionet.org`.

4. Human subjects. Does the research involve human subjects? **Not directly.** MedAlign data were collected with IRB approval by the original authors. No new human data collection was performed in this study. The physician adjudication of model responses involves clinicians reviewing de-identified examples and providing annotations, which constitutes professional quality assessment rather than human subjects research.

5. Risks. Are there any risks of harm from this research? **Yes; we discuss mitigations.** The findings could theoretically be used to identify clinical AI systems' performance gaps, but the intended application is to improve deployment safety. The clinical implications section explicitly cautions against deploying systems without hop-stratified evaluation.

B Hop Annotation Protocol

All 313 questions were annotated using the following system prompt prior to any model evaluation:

> *"You are annotating clinical EHR questions by their compositional reasoning depth (hop count). Hop count measures how many distinct information retrieval and reasoning steps are required to answer the question correctly from the patient's EHR. Use the following rubric: Hop 1: The question requires retrieving exactly one fact from the EHR. Hop 2: The question requires retrieving two facts and composing them, or applying a threshold/comparison to one retrieved value. Hop 3: The question requires three sequential reasoning steps, such*

as retrieving a fact, applying clinical knowledge to contextualize it, and then making a comparison or recommendation. Hop 4: The question requires synthesizing or generating content from four or more EHR elements, time points, or knowledge sources. Respond with only a JSON object: {'hop': 1|2|3|4, 'rationale': '...', 'ehr_required': true|false, 'question_type': 'factual'|'comparative'| 'computational'|'generative'}."

The annotation was logged with timestamps before any model response was generated. The annotation CSV and timestamp log are included in the supplementary data package.

Inter-rater reliability. A second independent annotation pass was run on a stratified random sample of 50 questions (12–13 per hop level, random seed 42) using the same system prompt and `claude-sonnet-4-6` model. A cross-model validation used `gpt-4o` on the same 50 questions.

Table 4: Hop annotation inter-rater reliability ($n = 50$ stratified sample).

Comparison	κ	**95% CI**	**Exact %**	**Adjacent %**
Claude orig vs. 2nd-pass	0.787	[0.603, 0.970]	84%	98%
Claude orig vs. GPT-4o	0.464	[0.292, 0.637]	60%	88%
Claude 2nd-pass vs. GPT-4o	0.581	[0.401, 0.761]	70%	—

Within-model reproducibility was substantial ($\kappa = 0.787$; Landis–Koch "substantial" range: 0.61–0.80). The 98% adjacent agreement (within ± 1 hop) indicates that most disagreements are between adjacent categories rather than extreme misclassifications. Cross-model agreement was moderate ($\kappa = 0.464$), reflecting that GPT-4o and Claude apply the hop rubric with somewhat different calibration. This divergence suggests that the hop construct captures a meaningful but partially model-dependent dimension of question complexity, and future studies should standardize the annotator or report cross-model reliability explicitly.

C Judge Validation

Synthetic calibration. Prior to the main experiment, we validated the automated judge (`gpt-4o-mini` with binary correct/incorrect scoring) on five hand-crafted question-reference-response triples: (1) a response that exactly reproduced the reference answer (expected: correct); (2) a response that stated a fact directly contradicted by the EHR (expected: incorrect, hallucination); (3) a complete refusal to answer (expected: incorrect, omission); (4) a response with correct facts but incorrect clinical interpretation (expected: incorrect, reasoning_error); (5) a response with partial but substantially accurate content (expected: correct). The judge produced the expected output for all five cases with confidence ≥ 0.85, confirming that the binary schema and definitions are operable.

Physician adjudication. Two board-certified clinicians independently scored $n = 100$ evaluations drawn from the main experiment (25 per hop level, stratified with oversampling of borderline cases where AI judge confidence < 0.70). Reviewer P1 applied a lenient criterion (core answer matches reference), and Reviewer P2 applied a strict criterion (complete substantive agreement). Reviewers were blinded to model condition (zero-shot vs. extended thinking). Cohen's κ was computed with Fleiss standard errors and exact normal-approximation 95% confidence intervals.

Table 5: Physician adjudication results ($n = 100$). Target: $\kappa \geq 0.70$.

Comparison	κ	**95% CI**	**Pass $\geq$0.70?**
AI judge vs. P2 (strict)	0.957	[0.874, 1.000]	✓
AI judge vs. P1 (lenient)	0.572	[0.364, 0.780]	×
P1 vs. P2 (inter-physician)	0.559	[0.345, 0.773]	—

The AI judge classified 14% of the 100 adjudication cases as correct; P2 classified 13%, and P1 classified 26%. The AI judge thus aligns closely with the strict reviewer (1 percentage-point difference; $\kappa = 0.957$, near-perfect by Landis and Koch conventions). Agreement with the lenient reviewer is moderate ($\kappa = 0.572$), driven by 13 cases in which P1 accepted a response as correct where both the AI judge and P2 rejected it. The single AI false positive was Case 89, in which the

model cited one of five required cancer screening recommendations; P1 credited this as partially correct but P2 and the AI judge scored it incorrect for omitting the four remaining recommendations.

The moderate inter-physician κ (0.559, 95% CI [0.345, 0.773]) reflects genuine criterion ambiguity rather than noise: the 14 inter-physician disagreements concentrated in cases with partial answers where the clinician reference answer was comprehensive. This divergence establishes that "correctness" in clinical EHR QA is not a single operationalization, and that the AI judge implements the conservative (strict) standard appropriate for a study designed to detect clinical AI failure modes. We treat $\kappa(\text{AI}, \text{P2}) = 0.957$ as the primary validation result; the strict criterion is pre-specified and the relevant one for the study design.

D Question-Type Stratification

Table 6: Zero-shot accuracy (%) by hop count and reasoning type (n shown per cell). Cells with $n < 5$ are marked as "–" (insufficient power). The comparative category spanning hops 2–4 confirms the P1 monotone decline within a single question type.

Reasoning type	**Hop=1**	**Hop=2**	**Hop=3**	**Hop=4**
Factual	30.3% (n=108)	10.0% (n=20)	—	—
Comparative	—	50.0% (n=16)	26.7% (n=30)	20.0% (n=20)
Generative	—	—	—	17.5% (n=80)
Computational	—	—	—	—
All types	*30.6% (n=111)*	*28.3% (n=46)*	*21.4% (n=42)*	*17.3% (n=104)*

Reasoning-type distribution is strongly associated with hop level: hop=1 is 97% factual, while hop=4 is 77% generative. Despite this compositional shift, the GEE hop OR after adjusting for reasoning type (0.57, 95% CI [0.38, 0.84], $p = 0.005$) is stronger than the unadjusted estimate (0.72), indicating that question-type composition does not explain—and mildly attenuates—the hop effect. Within the comparative type, Cochran–Armitage $z = -1.90$ ($p = 0.029$) across hops 2–4 confirms P1 within a single question category.

E ET-16K Selection Bias Analysis

Table 7: ET-16K completion rates and selection bias by hop level. "ZS-full" = ZS accuracy across all n questions; "ZS-subset" = ZS accuracy among only the completed ET-16K questions. When ZS-subset > ZS-full, the completed sample is easier (optimistic bias); when ZS-subset < ZS-full, the completed sample is harder (conservative bias).

Hop	**Completed**	**Timeout%**	**ZS-full**	**ZS-subset**	**ET-16K**	**Bias**
1	74/111	33.3%	30.6%	25.7%	28.4%	Conservative
2	32/46	30.4%	28.3%	25.0%	28.1%	Conservative
3	17/42	59.5%	21.4%	29.4%	29.4%	**Optimistic**
4	56/104	46.2%	17.3%	15.5%	19.6%	Conservative

The critical hop=3 result—which showed the highest timeout rate (59.5%)—used an optimistic subset: the completed questions had *higher* ZS accuracy (29.4%) than the full hop=3 sample (21.4%), indicating the harder questions timed out. Despite this favorable selection, ET-16K accuracy at hop=3 (29.4%) exactly equalled the ZS accuracy for the same subset (29.4%), yielding an ET benefit of exactly 0 pp. Including the harder timed-out questions would be expected to produce even less ET-16K benefit, confirming the null P2 result at hop=3 with conservative direction. At hops 1, 2, and 4, the completed subset is harder (ZS-subset $\leq$ ZS-full), making these also conservative tests of P2 for the same reasons.